David Khachatryan - Week 4 Report

Summary

Milestone 2 graph displays the P control of the pitch. As you can see the IMU pitch is constantly oscillating about zero. This implies that P is persistently trying to correct an error. As it never settles, it highlights the fact that the P controller is limited in this way. In addition, motor inputs are perfectly mirroring each other, indicating that there is pitch correcting behavior.

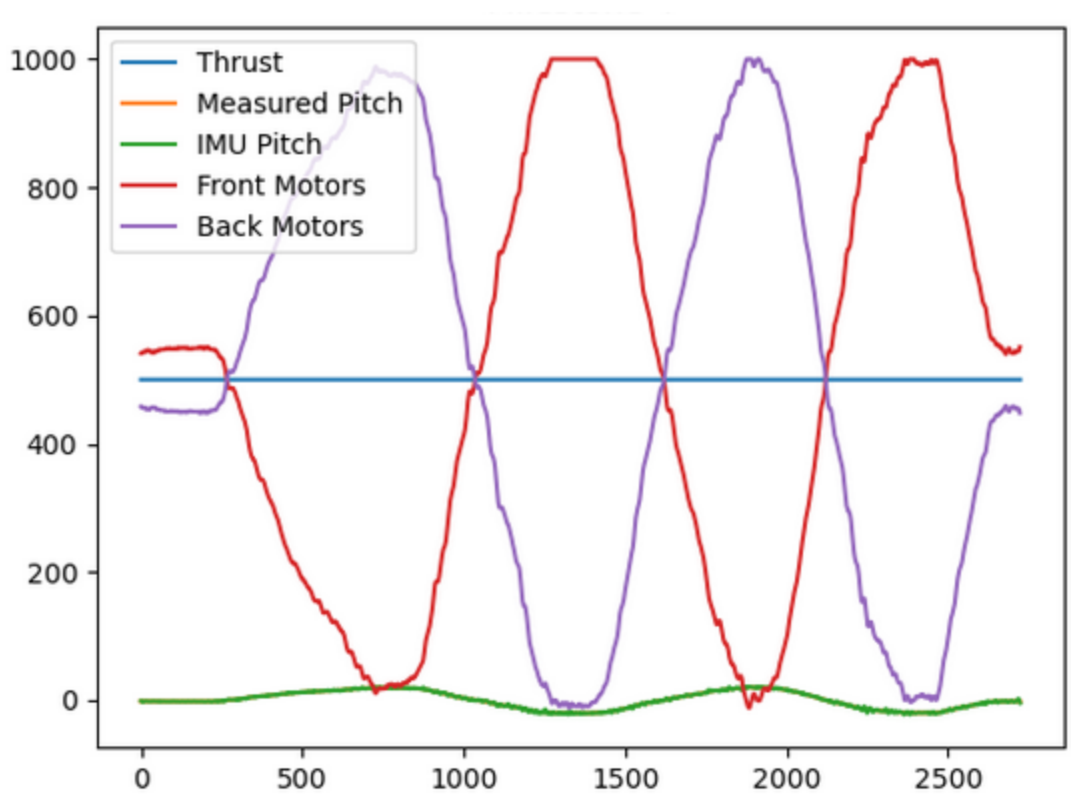

Figure 1: Milestone 2 Graph

In milestone 3 we implemented D control with the gain value of 1. The graph below shows the back motors going faster when gyro speed is going up. This movement creates the damping since the motors are creating force that goes against the pitch velocity.

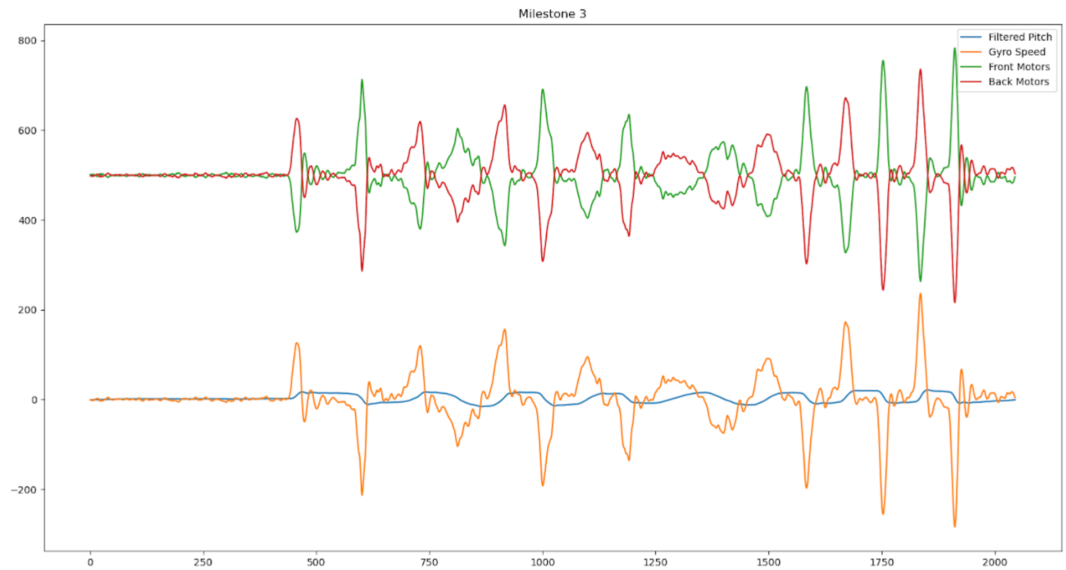

Figure 2: Milestone 3 Graph

In milestone 4 we tuned the P value to 30 and D value of 3.2. This still has some oscillations, so it might be underdamped. We would need to tune this further once we start flying.

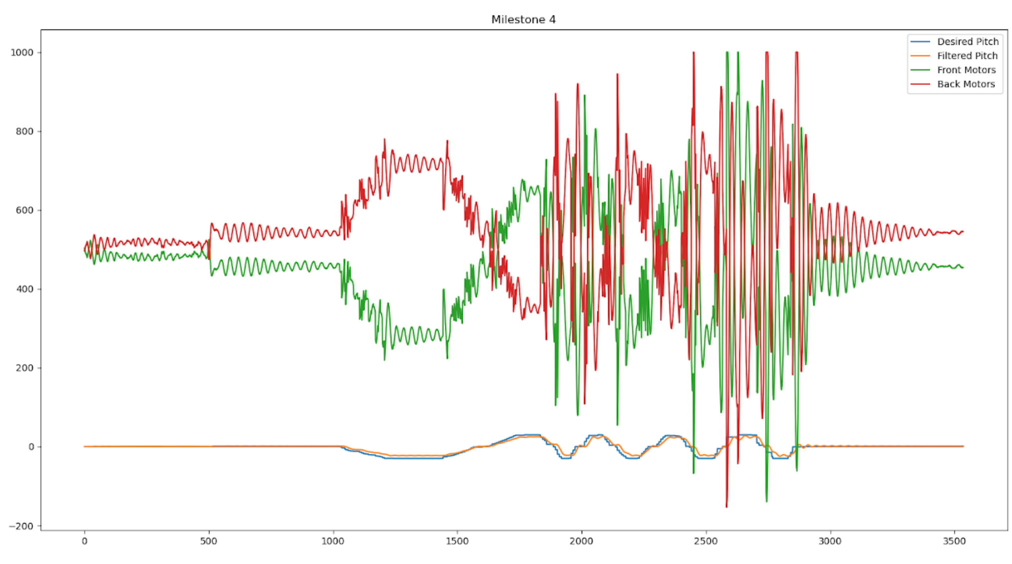

Figure 3: Milestone 4 Graph

In milestone 5, we changed the value in register 0x40 to 0x89, 0x99, and 0xA9. We had to increase the neutral thrust to create more noise. The 3 graphs below are sorted from least noise to most noise.

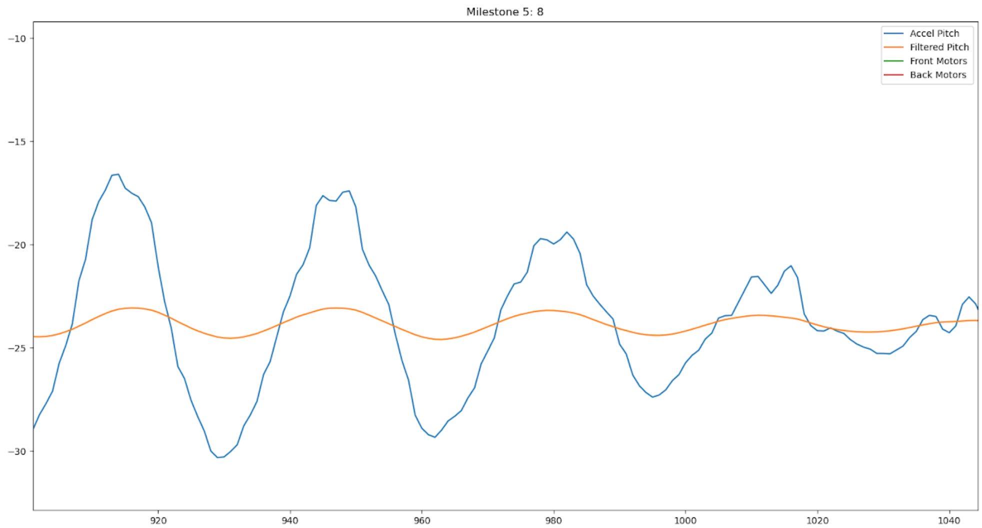

Figure 4: 4-fold oversampling

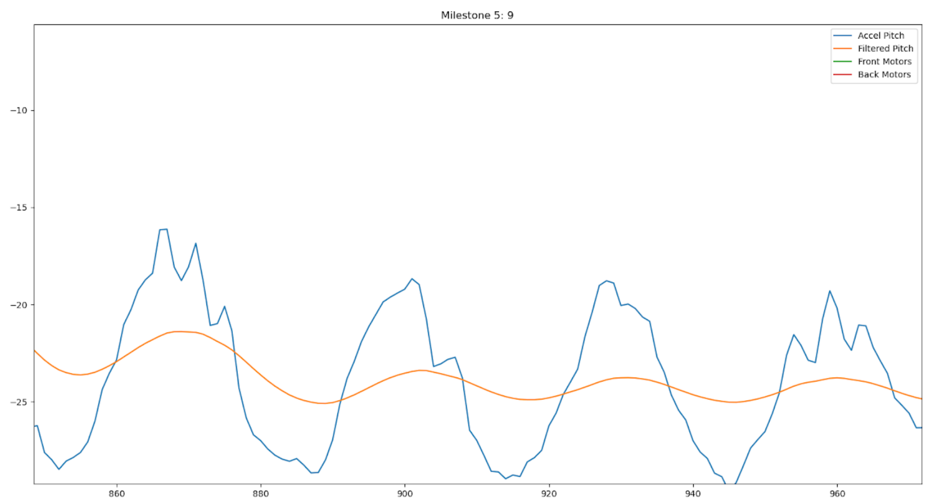

Figure 5: 2-fold oversampling

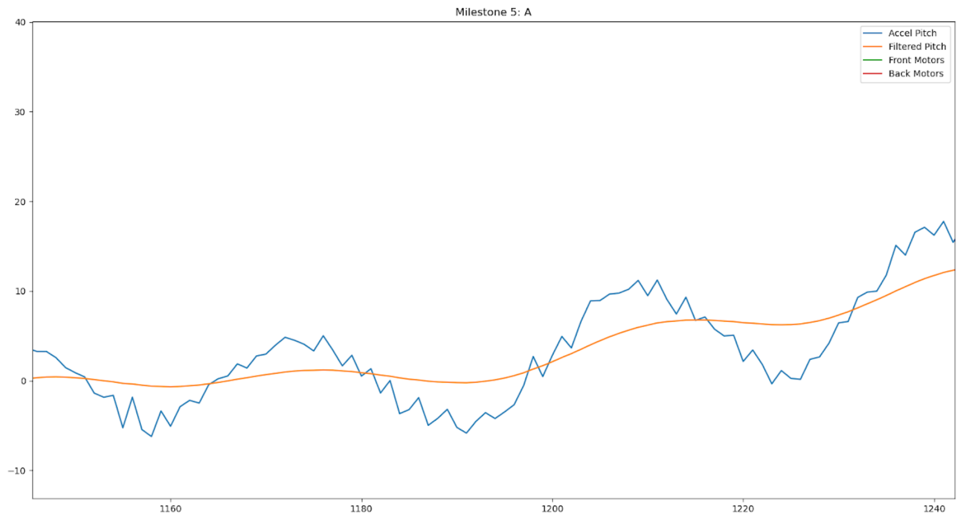

Figure 6: Normal

What went well

We got the PD controller up and running very quickly. Our tuning went well and it was not long enough until the damping was noticeable and the P and D gains were performing correctly.

What did not go well, why?

For milestone 5 we could not see the noise for different filtering values. We started by printing what was in the 0x40 register after we set the values 89, 99, and A9. The I2C print helper function confirmed that we were setting the right values to the register. After that, we increased the thrust neutral value. The motors were spinning faster, which created noise that was easier to distinguish.

What will you change for next class

We started working on a script to automate the startup of the udp listener and sender. We spent about 10 minutes last time debugging the controller, when we did not notice that the udp_rx listener binary was not running on the pi.

Contributions

We contributed equally during this week. During the entire process we worked together, asking questions and landing on the correct answer. Both of us were caught up with the codebase and fully understood the process.